Gaitrie Maleshia Motilall

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EDUCATION

New York Institute of Technology – Manhattan, NY – Graduated May 2026

- B.S. in **Electrical and Computer Engineering** || Minor in **Mathematics** || **3.64 GPA** || Honors List

Relevant Coursework: Electrical Circuits II, Microprocessors, Electronics II, Antenna Theory, Communication Systems.

TECHNICAL SKILLS & CERTIFICATIONS

- **Software:** I²C, MATLAB, PSpice, Altair FEKO, KiCad, Python, C++, Java, Microsoft Office, GitHub, AutoCAD.
- **Hardware:** Soldering, Power & Grounding, Current Sizing, Electromechanical Systems, Motor Drives, Analog & Mixed Signal System Design, Arduino Uno, PSoC3, Timing & Data Acquisition.
- **Certifications:** OSHA 30-Hour Construction Training (Completed), NFPA 70E (Completed), Drivers License.

PROJECTS

Microcontroller Based Railway Direction Control System

- Drafted PCB schematics and layouts in KiCad for a Raspberry Pi CM4-based railway control system, integrating board-to-board connectors, USB-C connectivity, and embedded control interfaces.
- Allocated GPIO interfaces for five directional pushbuttons, five corresponding LED indicator circuits, and accessible test points to support signal verification and troubleshooting.
- Optimized component placement and electrical routing for through-hole LEDs and an I²C OLED display within a unified PCB design.

Technology Used: KiCad, Raspberry Pi CM4, USB-C, GPIO, 5-Position Rotary Selector, Through Hole LEDs, I²C, OLED Display.

Simulation of a Top Hat Infinitesimal Dipole Antenna

- Modeled a top-hat infinitesimal dipole antenna in FEKO to study its RF radiation characteristics at 30 MHz.
- Configured electromagnetic simulations using a 1 V excitation source, PEC conductors, defined antenna geometry, and a fine mesh to evaluate far-field radiation patterns, directivity, and radiation resistance.
- Validated simulated antenna performance against theoretical calculations, achieving directivity within 0.20 dB of the expected value and a 10.97% difference in radiation resistance.

Senior Design – NYIT - Embedded Electrical Impedance Spectroscopy System for Skin Cancer Detection

- Led the design and implementation of a noninvasive impedance measurement system with controlled electrode switching and real-time data acquisition across multiple probe configurations.
- Developed embedded control and communication systems for automated frequency sweeps and continuous impedance data logging.
- Engineered a low-noise signal path using a custom guard-shield circuit to improve signal integrity.
- Designed and fabricated a custom scanning enclosure and probe system, integrating actuated motion control with calibration and hardware validation.

Technology Used: AD5933, Arduino Uno, CD4053BE, OPA376 (SC70-5), NEMA 17 Stepper Motor, A4988 Driver, I2C, Blender, Shapr3D, Solder, Breadboarding, Electrical Testing (Multimeter), Github, GUI.

WORK EXPERIENCE

NSF – New York NY - Research Assistant – Electromagnetic & Impedance Research –May 2023 to July 2024

- Conducted simulation-driven EIS research by modeling biological tissue and analyzing impedance response to evaluate electrode geometry, frequency variation, and tissue properties for cancer detection.
- Investigated low-frequency electromagnetic and microwave imaging systems, supporting development of noninvasive diagnostic methodologies.
- Analyzed experimental and simulated datasets to compare tissue responses, informing system design decisions and research direction; documented findings in a formal research paper.

Technologies Used: MATLAB, FEKO, Electromagnetic Simulation, Data Analysis.

Mars (M&Ms) – New York NY - Technical Operations Associate –February 2023 to October 2025

- Operated, maintained, and troubleshot high-throughput electromechanical printing systems, reducing downtime by ~15% through fault isolation, mechanical adjustment, and system recovery.
- Diagnosed and resolved failures across mechanical subsystems (rollers, feed mechanisms) and control interfaces, ensuring reliable operation in continuous production environments.
- Disassembled, cleaned, and reassembled precision machine components, restoring mechanical alignment, electrical reliability, and system performance through preventative component servicing.